

AI Exam Prep — Master Study Index

Reading time (this page): ~10 min as orientation; the *Analogies Index* and *Pitfalls Compendium* are the night-before-the-exam quick scans. **Scope:** 10 lecture chapters + 11 lab solutions + shared glossary + cross-reference graph. **How this index is organised:** §1 orientation, §2 study order, §3 full TOC, §4 analogies (this is the most valuable section — start here when reviewing), §5 flattened glossary, §6 cross-reference graph, §7 lab reference table, §8 pitfalls compendium.

1. What this package is and how to use it

This is a self-contained study package for the *Introduction to Artificial Intelligence* course. For each of the 10 lecture topics the package contains:

- **A chapter** `.md` in `lectures/` with eight sections — Overview, Big-Picture Analogies, Core Concepts, Algorithms/Methods, Worked Examples, Common Pitfalls / Exam Traps, Connections to Other Lectures, and a Cheat-Sheet Summary. Each chapter renders to a printable `.pdf` of the same name.
- **A corresponding lab** with a working `*_solution.py` (or `.ipynb`) sitting alongside the original handout template, plus a `variants.md` of exam-style variant prompts under `_exam/{LabName}/`.

Three usage modes the package is designed for:

1. **Studying a topic from scratch.** Open the chapter `.md`. Read §1, §2 (analogies first — they give you the mental model), then §3/§4 for rigour, §5 for worked numbers. Skim §6 last.
2. **Reviewing for the exam.** Open this index. Scan §4 Analogies, then §8 Pitfalls. Drill any concept whose analogy doesn't ring a bell by jumping to its lecture §3 entry.
3. **Practising labs.** Open the relevant lab template, attempt it, then diff against `*_solution.py`. Read the lab's `variants.md` for exam-style perturbations.

All math uses LaTeX delimiters (rendered to KaTeX on PDF). All links are relative paths from `study/`.

2. Recommended Study Order

The lecture sequence below matches the course flow. Reading times are estimates for first-pass study; expect to halve them on review passes.

#	Chapter	First-pass time	What you walk away knowing
1	L02 — Agents	60–75 min	Agent / environment / PEAS / 6 environment dimensions / 5 agent architectures. <i>Vocabulary for every later lecture.</i>
2	L03 — Uninformed Search	45–60 min	BFS, DFS, UCS, IDS; the search-tree mechanism (frontier, expansion, completeness, optimality).
3	L05 — Local Search	60–75 min	Hill climbing, simulated annealing, genetic algorithms; objective function vs heuristic.
4	L06 — Adversarial Search	45–60 min	Game tree, minimax, alpha-beta pruning, evaluation functions.
5	L07 — CSP	60 min	Variables/domains/constraints; backtracking + MRV/degree/LCV + forward checking + arc consistency.
6	L09a — Bayesian Networks	75–90 min	Random variables, joint/marginal/conditional, Bayes' rule, BN structure, Markov condition, inference by enumeration, Naive Bayes.
7	L09b — HMM	60–75 min	Markov chains, HMM $\lambda = (A, B, \pi)$, three problems (Evaluation/Decoding/Learning), forward and Viterbi recursions.
8	L10 — Intro to ML	60 min	Supervised/unsupervised/RL split; decision trees, Gini/entropy/info-gain, overfitting and pruning.
9	L11 — Regression	60–75 min	Linear regression, OLS, residuals, SST/SSE/SSR, R^2 , p-values, dummy variables, interaction terms, multicollinearity.
10	L12 — Clustering	45–60 min	K-means, hierarchical (5 linkages), DBSCAN; centroid, dendrogram.

Total first-pass time: ~10–13 hours of reading. Add ~1 hour per lab if you also re-implement them.

"If you only have one day"

Read in this priority order — stop when the clock runs out:

1. §4 [Analogies Index](#) below (1 h) — install the mental models.
2. §8 [Common Pitfalls Compendium](#) (1 h) — pre-immunise against the exam traps.
3. The Cheat-Sheet Summary (§8) of each lecture, in study order (2 h).
4. §5 [Glossary](#) — skim, focus on terms you can't define from memory (1 h).
5. The pitfalls section (§6) of any lecture whose mental model still feels shaky (1–2 h).

"If you have a week"

Day	Plan
1	L02 + L03 chapters, work Lab 1 (Agents) and Lab 2 (Search).
2	L05 chapter, work Lab 4 (GA).
3	L06 + L07 chapters, work Lab 5 (Alpha-Beta) and Lab 6 (CSP).
4	L09a + L09b chapters, work Lab 7 (BN) and Lab 8 (HMM).
5	L10 + ML Lab 1 (Classification).
6	L11 + ML Lab 2 (Regression); L12 + ML Lab 3 (Clustering).
7	Re-read §4 Analogies + §8 Pitfalls below; revisit every chapter's Cheat-Sheet Summary; attempt variant prompts in <code>_exam/*/variants.md</code> .

"If you have a month"

Add a second pass per chapter — read §5 Worked Examples in detail and re-derive every result with pencil and paper. Then attempt each `variants.md` cold (no solution open) and self-mark.

3. Full Table of Contents

Every lecture, every major heading, every named concept. Click to jump.

L02 — Introduction to Agents

- 1. Overview & Motivation
- 2. The Big Picture — Analogies
- 3. Core Concepts
 - 3.1 Agent and environment
 - 3.2 Percept, percept sequence, agent function, agent program
 - 3.3 Rationality and performance measure
 - 3.4 Autonomy
 - 3.5 PEAS
 - 3.6 Environment types (the six-dimensional taxonomy)
 - 3.7 Hierarchy of agent types
- 4. Algorithms / Methods
 - 4.1 Table-driven agents
 - 4.2 Simple reflex agent
 - 4.3 Model-based reflex agent
 - 4.4 Goal-based agent
 - 4.5 Utility-based agent
 - 4.6 Learning / autonomous agent
- 5. Worked Examples

- [6. Common Pitfalls / Exam Traps](#)
- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L03 — Uninformed Search](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Problem-solving agent](#)
 - [3.2 Search problem definition](#)
 - [3.3 State, node, state space](#)
 - [3.4 Search tree, frontier, node expansion](#)
 - [3.5 Branching factor, depth, max depth](#)
 - [3.6 Search-strategy evaluation dimensions](#)
 - [3.7 Uninformed vs informed search](#)
 - [3.8 A* \(forward reference only\)](#)
- [4. Algorithms / Methods](#)
 - [4.1 Breadth-first search \(BFS\)](#)
 - [4.2 Uniform-cost search \(UCS\)](#)
 - [4.3 Depth-first search \(DFS\)](#)
 - [4.4 Iterative deepening search \(IDS\)](#)
 - [4.5 Side-by-side comparison](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L05 — Local Search](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Objective function](#)
 - [3.2 State-space landscape](#)
 - [3.3 Hill climbing \(greedy local search\)](#)
 - [3.4 Variants — escaping local maxima](#)
 - [3.5 Simulated annealing](#)
 - [3.6 Population-based search and genetic algorithms](#)
 - [3.7 Fitness landscapes](#)
- [4. Algorithms / Methods](#)
 - [4.1 Hill climbing](#)
 - [4.2 Simulated annealing](#)

- [4.3 Genetic algorithm](#)
- [4.4 Comparison table](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
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- [8. Cheat-Sheet Summary](#)

[L06 — Adversarial Search](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Games as a multi-agent task environment](#)
 - [3.2 Game tree, MAX, MIN, terminal states](#)
 - [3.3 Minimax value of a state](#)
 - [3.4 Evaluation function](#)
 - [3.5 Zero-sum and the MAX / MIN labelling](#)
- [4. Algorithms / Methods](#)
 - [4.1 The Minimax algorithm](#)
 - [4.2 Alpha-beta pruning](#)
 - [4.3 Properties of alpha-beta](#)
 - [4.4 Comparison table](#)
 - [4.5 Additional techniques](#)
 - [4.6 Games of chance — expectiminimax](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L07 — Constraint Satisfaction Problems](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Variable, Domain, Constraint](#)
 - [3.2 CSP as a Search Problem](#)
 - [3.3 Constraint graph](#)
 - [3.4 Example domains](#)
 - [3.5 Why naive search is bad here](#)
- [4. Algorithms / Methods](#)
 - [4.1 Backtracking search](#)
 - [4.4 MRV and degree heuristic](#)
 - [4.5 LCV value ordering](#)

- 4.6 Forward checking
- 4.8 Arc consistency / AC-3
- 5. Worked Examples
- 6. Common Pitfalls / Exam Traps
- 7. Connections to Other Lectures
- 8. Cheat-Sheet Summary

L09a — Bayesian Networks

- 1. Overview & Motivation
- 2. The Big Picture — Analogies
- 3. Core Concepts
 - 3.1 Random variables and events
 - 3.2 Atomic events, joint distribution, marginal
 - 3.3 Conditional probability, Bayes' rule, normalisation
 - 3.5 Independence
 - 3.6 Conditional independence
 - 3.7 Bayesian network — definition
 - 3.8 Conditional probability table (CPT)
 - 3.9 The Markov condition
 - 3.10 The joint distribution from a Bayesian network
 - 3.11 Naive Bayes classifier
 - 3.13 Inference by enumeration
 - 3.14 Uncertainty, prior, posterior, evidence
- 4. Algorithms / Methods
- 5. Worked Examples
- 6. Common Pitfalls / Exam Traps
- 7. Connections to Other Lectures
- 8. Cheat-Sheet Summary

L09b — Hidden Markov Models

- 1. Overview & Motivation
- 2. The Big Picture — Analogies
- 3. Core Concepts
 - 3.1 The HMM picture
 - 3.2 Markov chains — the observable foundation
 - 3.3 Hidden states and observations — the HMM upgrade
 - 3.4 The Fair Bet Casino
 - 3.5 The three basic problems
- 4. Algorithms / Methods
 - 4.1 The naive approach (and why it fails)
 - 4.2 The forward algorithm

- [4.3 The Viterbi algorithm](#)
- [4.4 Forward vs Viterbi](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L10 — Introduction to Machine Learning](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Three branches of machine learning](#)
 - [3.2 Supervised learning](#)
 - [3.3 Classification, formally](#)
 - [3.4 Common classification techniques](#)
 - [3.5 Decision tree \(informal\)](#)
- [4. Algorithms / Methods](#)
 - [4.2 Tree induction algorithms](#)
 - [4.4 Best split — impurity framework](#)
 - [4.5 Gini impurity](#)
 - [4.6 Entropy](#)
 - [4.7 Classification error](#)
 - [4.8 Stopping criteria](#)
 - [4.9 Ensemble methods](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L11 — Regression](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 What is a model? Predictive modelling](#)
 - [3.2 Linear regression — the simple case](#)
 - [3.3 Residuals and OLS](#)
 - [3.4 Interpreting intercept and slope](#)
 - [3.5 Sum-of-squares decomposition: \$SST = SSR + SSE\$](#)
 - [3.6 R-squared \(\$R^2\$ \)](#)
 - [3.7 Adjusted \$R^2\$](#)
 - [3.8 p-value](#)

- [3.9 Confidence intervals](#)
- [3.10 Dummy variables](#)
- [3.11 Interaction terms](#)
- [3.12 Multicollinearity](#)
- [3.13 Why linearity is restrictive](#)
- [4. Algorithms / Methods](#)
- [5. Worked Examples](#)
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- [7. Connections to Other Lectures](#)
- [8. Cheat-Sheet Summary](#)

[L12 — Clustering](#)

- [1. Overview & Motivation](#)
- [2. The Big Picture — Analogies](#)
- [3. Core Concepts](#)
 - [3.1 Cluster analysis \(the task\)](#)
 - [3.2 Types of clustering](#)
 - [3.3 Centroid](#)
 - [3.4 Dendrogram](#)
 - [3.5 Hierarchical clustering](#)
 - [3.6 Agglomerative clustering](#)
 - [3.7 DBSCAN — density-based clustering](#)
- [4. Algorithms / Methods](#)
 - [4.1 K-means](#)
 - [4.2 K-means issues and limitations](#)
 - [4.4 Hierarchical linkages](#)
 - [4.5 Bisecting K-means](#)
 - [4.6 DBSCAN algorithm](#)
 - [4.7 Algorithm comparison](#)
- [5. Worked Examples](#)
- [6. Common Pitfalls / Exam Traps](#)
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4. Analogies Index

This is **the most valuable single page** in this package. When you're reviewing for the exam, scan this table first — if the analogy doesn't immediately bring the mental model to mind, jump to the lecture's §2 entry and re-read its caveat.

Every analogy below has a "where it breaks down" caveat in the source — mis-applied analogies are the single most common source of exam mistakes (e.g. confusing "deterministic" with "predictable", confusing "filtering" between slide-speak and textbook-speak). Follow the link if the analogy doesn't quite land.

Concept	One-line analogy	Source
Agent	A thermostat with a job description — senses, decides, acts in a loop.	L02 §2
Agent function vs program	The contract vs the employee — function is the spec, program is the code.	L02 §2
Rational agent	A poker player who plays expected value — best decision given evidence, not best outcome.	L02 §2
Percept vs percept sequence	One movie frame vs the whole movie up to now.	L02 §2
Performance measure	The referee at a football match — judges by the goals, not the effort.	L02 §2
PEAS	A one-page brief for a freelance contractor — Performance, Environment, Actuators, Sensors.	L02 §2
Environment types	Six switches on a job description — flip each one wrong and your agent fails predictably.	L02 §2
Table-driven agent	An infinite, impossible filing cabinet keyed by every life-history of the robot.	L02 §2
Simple reflex agent	A vending machine — press B-4, get a Mars bar; no memory.	L02 §2
Model-based reflex agent	A driver in fog — keep an internal model of what's out there.	L02 §2
Goal-based agent	A satnav — explicit destination, route is recomputed when the world changes.	L02 §2
Utility-based agent	A satnav with preferences — "avoid motorways, prefer scenic"; real-valued score.	L02 §2
Learning / autonomous agent	An apprentice with a coach, an internal update, and the nerve to try something new.	L02 §2
State-space search	Exploring a maze — start at the entrance, find the goal.	L03 §2
BFS	A postman knocking on every house on each street before moving one block out.	L03 §2
DFS	Always take the leftmost corridor; back up only at a wall.	L03 §2
UCS	Dijkstra's wavefront — water flooding the cheapest paths first.	L03 §2
IDS	Run BFS but pretend each new depth level is a fresh maze.	L03 §2
Frontier vs explored set	"People I plan to call" vs "people I've already called."	L03 §2
Local search	Hiking with only an altimeter, no map — only the final altitude matters.	L05 §2

Concept	One-line analogy	Source
Hill climbing	Always step uphill, blindfolded — Everest in fog with amnesia.	L05 §2
Local maximum	A small foothill — every direction goes down, so you stop.	L05 §2
Plateau	Wide flat patch — every direction looks equally flat.	L05 §2
Ridge	Narrow rocky chain — each step off looks worse but walking along gains altitude.	L05 §2
Random-restart hill climbing	If stuck on a foothill, helicopter to a random spot and climb again.	L05 §2
First-choice hill climbing	Take the <i>first</i> step that goes up — don't sniff every direction.	L05 §2
Stochastic hill climbing	Roll a die among the improving directions.	L05 §2
Simulated annealing	Shaking a settling marble — vigorous when hot, gentle when cool.	L05 §2
Temperature schedule	The dimmer-switch on the shaker.	L05 §2
Genetic algorithm	Animal breeding — population, selection, crossover, mutation, generations.	L05 §2
Chromosome / genotype	The DNA strand that is one candidate solution.	L05 §2
Fitness function	The breeder's eye — "how good is this puppy?"	L05 §2
Roulette-wheel selection	A casino wheel with slot widths rigged proportional to fitness.	L05 §2
Crossover	Swap the front and rear of two prototype cars to build a third.	L05 §2
Mutation	A one-letter typo when copying out a long word.	L05 §2
Elitism	Always keep one perfect copy of the best chromosome so far.	L05 §2
Fitness landscape	Topography of all chromosomes, elevation = fitness.	L05 §2
Minimax	You and a perfectly devious sibling take turns; you both look to the end of the game.	L06 §2
Alpha-beta pruning	"Stop reading a bad chess line the moment you realise it's worse than one you have."	L06 §2
α / β	α is the floor MAX has guaranteed; β is the ceiling MIN has imposed.	L06 §2
Evaluation function	A one-glance score of who's winning this chess position.	L06 §2
Zero-sum game	"The bigger I win, the bigger you lose, by exactly the same amount."	L06 §2

Concept	One-line analogy	Source
CSP	Filling a Sudoku grid — try a value, backtrack on conflict, repeat.	Lo7 §2
Constraint graph	A wedding seating chart — guests are nodes, "can't sit together" are edges.	Lo7 §2
MRV	Tackle the most-cornered Sudoku square first — fail fast.	Lo7 §2
LCV	Pick the value that leaves the most doors open for other variables.	Lo7 §2
Forward checking	When you place a queen, cross off her attacked squares on every future row's candidate list.	Lo7 §2
Arc consistency	A customs queue with reciprocal stamps — passports lose pages until every page has support.	Lo7 §2
Backtracking	Trying outfits before a wedding — swap the trousers, then the shirt, before abandoning.	Lo7 §2
Consistent assignment	A Tetris board mid-game — partial but no rule broken yet.	Lo7 §2
Degree heuristic	Seat the maiden aunt with the most feuds first.	Lo7 §2
Constraint propagation	Newton's cradle — one assignment ripples through the chain.	Lo7 §2
Random variable	A thermometer reading at a moment you cannot pre-specify.	Lo9a §2
Atomic event / joint distribution	One row of the world's master spreadsheet / the whole spreadsheet.	Lo9a §2
Marginal probability	Projecting the spreadsheet down to one column.	Lo9a §2
Conditional probability	Restricting the spreadsheet to one office, then asking about it.	Lo9a §2
Bayes' rule	Flipping the causal arrow to answer a diagnostic question.	Lo9a §2
Chain rule	Handing out blame for the joint, one variable at a time.	Lo9a §2
CPT	A child's behaviour chart on the fridge — one row per parent-mood combo.	Lo9a §2
Independence	Two questions that share no information — die roll vs coin flip.	Lo9a §2
Bayesian network	A gossip graph — each person depends only on their direct informants.	Lo9a §2
Markov condition	Parents acting as gatekeepers for upstream gossip.	Lo9a §2
Conditional independence	Once I know the rain, the cloud doesn't change my traffic prediction.	Lo9a §2

Concept	One-line analogy	Source
Prior / posterior / evidence	The weather forecast updating after the first lightning strike.	L09a §2
Inference by enumeration	Summing the entire phone book to count "everyone whose surname starts with S."	L09a §2
Naive Bayes	A gossip graph with one root and many leaves; the leaves don't talk to each other.	L09a §2
HMM	Watching the umbrella to guess the weather inside your neighbour's windowless house.	L09b §2
Markov chain	A board game where your next square depends only on your current square.	L09b §2
Forward algorithm	Totalling all the ways the story could have unfolded.	L09b §2
Viterbi	GPS with breadcrumbs — best route reconstructed from where I am now.	L09b §2
Forward vs Viterbi	Sum vs max — same trellis, one operator difference.	L09b §2
First-order Markov assumption	Only checking the latest weather report — forget the deeper history.	L09b §2
Initial distribution π	The climatological prior on the day you arrived in Reykjavik.	L09b §2
Filtering vs evaluation	"Is it raining right now?" vs "how likely was the whole week?"	L09b §2
Supervised learning	Studying with an answer key on the back of every flashcard.	L10 §2
Unsupervised learning	Sorting laundry without being told the categories.	L10 §2
Reinforcement learning	Learning a video game without reading the manual — score is the only signal.	L10 §2
Classification	Deciding "which bin does this letter go in?" at the post office.	L10 §2
Regression	Predicting tomorrow's temperature — a continuous number, not a label.	L10 §2
Decision tree	A game of 20 questions — each internal node asks one, each leaf is the answer.	L10 §2
Random forest	Asking many slightly-different experts and taking the majority vote.	L10 §2
Overfitting	Memorising past-paper answers instead of understanding the topic.	L10 §2

Concept	One-line analogy	Source
Gini / entropy / classification error	"How messy is this drawer?" — three slightly different mess metrics.	L10 §2
Linear regression	Drawing one straight flight path on a 2-D chart of cities.	L11 §2
Residual	How far each city sits above or below the flight path.	L11 §2
OLS	Choose the flight path minimising the sum of squared city-offsets.	L11 §2
R^2	The share of the data's spread that the flight path absorbs.	L11 §2
p-value	Probability the apparent effect is random noise dressed up to look real.	L11 §2
Confidence interval	A dart-throw target ring around the true slope.	L11 §2
Dummy variable	A light switch wired to a fixed bonus inside the equation.	L11 §2
Interaction term	A throttle on the slope — engaged by the dummy, sized by the interaction coef.	L11 §2
Multicollinearity	Two passports for the same person — algorithm can't tell which one is "doing the work".	L11 §2
Cluster analysis	Sorting laundry without being told the categories.	L12 §2
K-means	K party hosts each claiming the nearest guests, then shuffling to the centre of their circle.	L12 §2
Hierarchical clustering	Building a family tree of point similarities by repeated "marriages".	L12 §2
Centroid	The gravitational centre of a cluster — average of its members' coordinates.	L12 §2
DBSCAN	Wander through dense neighbourhoods, leave the loners behind.	L12 §2
Dendrogram	An upside-down family tree; cut horizontally at any height to get flat clusters.	L12 §2

73 analogies across 10 lectures. Each one is the *first* thing to scan when reviewing a topic; the lecture's §2 entry adds the "where it breaks down" caveat that prevents over-extrapolation.

5. Glossary (flattened)

Every glossary entry, one-line definition, link to the introducing lecture's §3 section. Full multi-paragraph definitions, notation, and reuse-history live in `_shared/glossary.md`.

Term	One-line definition	Introduced in
A* search	Best-first informed search expanding the node minimising $f(n) = g(n) + h(n)$. With admissible/consistent h , complete and optimal.	Lo3 §3.8 (FWD-REF)
AC-3 (arc-consistency algorithm)	Worklist algorithm that enforces arc consistency by repeatedly re-queueing arcs whose source domain has shrunk.	Lo7 §4.8
Action	A discrete choice available to an agent; an operator transforming a state into a successor.	Lo2 §3.1
Adversarial search	Search in environments with goal-conflicting agents, modelled as two-player zero-sum games.	Lo6 §3.1
Agent	Anything that perceives its environment through sensors and acts via actuators.	Lo2 §3.1
Agent function	Mathematical mapping $f: \mathcal{P}^* \rightarrow A$ from percept sequences to actions.	Lo2 §3.2
Agent program	Concrete code (on the architecture) that realises the agent function.	Lo2 §3.2
Agglomerative clustering	Bottom-up hierarchical clustering: start with singletons, merge closest pair, repeat.	L12 §3.6
Alpha cutoff	Pruning at a MIN node when its running β drops to or below an ancestor MAX's α .	Lo6 §4.2
Alpha-beta pruning	Minimax optimisation: prune subtrees that can't affect the root; best-case $O(b^{d/2})$.	Lo6 §4.2
Arc consistency	Binary CSP arc $X \rightarrow Y$ is consistent iff every $x \in D_X$ has a supporting $y \in D_Y$.	Lo7 §4.8
Atomic event	A complete value assignment to every random variable; one cell of the full joint.	Lo9a §3.2
Autonomy (autonomous agent)	Property of an agent whose behaviour is shaped by experience, not just by built-in knowledge.	Lo2 §3.4
Backtracking search (CSP)	DFS adapted to CSPs: assign one variable, back up the moment a constraint is violated.	Lo7 §4.1
Bagging	Ensemble: many models on bootstrapped subsets, averaged/voted to reduce variance.	L10 §4.9
Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$; flips causal into diagnostic.	Lo9a §3.3
Bayesian network		Lo9a §3.7

Term	One-line definition	Introduced in
	DAG of random variables, edges = direct dependences, each node carries $P(X_i \mid \text{Pa}(X_i))$.	
Beta cutoff	Pruning at a MAX node when its α rises to or above an ancestor MIN's β .	L06 §4.2
Branching factor	Max successors of any node in a search tree, b . Together with depth: tree size $O(b^d)$.	L03 §3.5
Breadth-first search (BFS)	Expand the shallowest unexpanded node (FIFO queue). Complete; optimal for equal step costs.	L03 §4.1
Centroid	Mean of points in a cluster — K-means update target each iteration.	L12 §3.3
Chain rule (probability)	$P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$; BNs collapse it to parents only.	L09a §3.3
Chromosome (GA)	Encoded representation of a candidate solution (classically a fixed-length bitstring).	L05 §3.6
Classification	Supervised learning with discrete labels (vs regression's continuous outputs).	L10 §3.3
Clustering	Unsupervised partitioning so intra-cluster distance is small, inter-cluster distance large.	L10 §3.1
Completeness	Algorithm guaranteed to find a solution when one exists (and report failure otherwise).	L03 §3.6
Conditional independence	$X \perp Y \mid Z$: $P(X, Y \mid Z) = P(X \mid Z) P(Y \mid Z)$. Structural axiom of BNs.	L09a §3.6
Conditional probability	$P(A \mid B) = P(A \cap B) / P(B)$; updating belief given evidence.	L09a §3.3
Conditional probability table (CPT)	Table of $P(X_i \mid \text{Pa}(X_i))$ for every parent-value combination.	L09a §3.8
Consistent assignment (CSP)	Partial assignment that violates no constraint (may still be incomplete).	L07 §3.1
Constraint	Restriction on which combinations of values are allowed for a subset of CSP variables.	L07 §3.1
Constraint graph	Nodes = CSP variables; edges = variables co-occurring in a binary constraint.	L07 §3.3
Constraint propagation	Locally enforcing consistency to prune domains before/during backtracking.	L07 §4.6
		L07 §3.1

Term	One-line definition	Introduced in
Constraint Satisfaction Problem (CSP)	$\langle X, D, C \rangle$ — variables, domains, constraints; solution is a complete consistent assignment.	
Crossover	GA recombination: combine two parent chromosomes (typically by cut-and-swap) into offspring.	L05 §3.6
d-separation	Graphical criterion on a BN for conditional independence of variable sets.	L09a §3.9
DBSCAN	Density-based clustering: core / border / noise points; clusters are density-connected core sets.	L12 §3.7
Decision tree	Tree classifier: internal nodes test attributes, branches are values, leaves carry labels.	L10 §3.5
Degree heuristic	CSP tie-breaker (within MRV): pick variable with most edges to unassigned neighbours.	L07 §4.4
Dendrogram	Tree diagram recording merges/splits of hierarchical clustering.	L12 §3.4
Depth-first search (DFS)	Expand the deepest unexpanded node (LIFO stack). Not complete on infinite/cyclic; $O(bm)$ space.	L03 §4.3
Deterministic vs stochastic	Whether the next state is fully determined by current state + action.	L02 §3.6
Discrete vs continuous	Whether percepts/actions/states/time are countable or take values in a continuum.	L02 §3.6
Dummy variable	Binary $\{0,1\}$ recoding of a categorical attribute for use in numeric regression.	L11 §3.10
Elitism	GA selection variant: best chromosome(s) copied unchanged into the next generation.	L05 §3.6
Emission model (observation model, HMM)	$b_k(o) = P(o \mid q = k)$; row k of matrix B in $\lambda = (A, B, \pi)$.	L09b §3.3
Ensemble method	Combine many base learners (vote or average) — bagging, boosting, random forest.	L10 §4.9
Entropy (splitting criterion)	$\text{Entropy}(t) = -\sum_j p(j \mid t) \log_2 p(j \mid t)$; impurity measure for ID3/C4.5.	L10 §4.6
Environment	Everything outside the agent — what it perceives and acts upon.	L02 §3.1
Environment types (taxonomy)	Six binary axes: observable, deterministic, episodic, static, discrete, single-agent.	L02 §3.6

Term	One-line definition	Introduced in
Episodic vs sequential	Whether the choice in one episode affects future episodes.	Lo2 §3.6
Evaluation function	Heuristic estimate of utility at a non-terminal game state, used at the depth cutoff.	Lo6 §3.4
Evidence (Bayesian)	Observed values of variables we condition on; E in queries $P(X \mid E)$.	Lo9a §3.14
Expected utility	$\sum_i P(\text{outcome}_i \mid \text{action}) U(\text{outcome}_i)$; rational agents maximise it.	Lo9a §3.3
Filtering (HMM Problem 1)	Computing $P(O \mid \lambda)$ — likelihood of the observation sequence; solved by forward algorithm.	Lo9b §3.5
Fitness function	Real-valued GA score of how good a chromosome is for the problem.	Lo5 §3.6
Forward algorithm (HMM)	DP that fills $\alpha_t(j) = P(o_1, \dots, o_t, q_t = j \mid \lambda)$ in $O(N^2 T)$.	Lo9b §4.2
Forward checking	After assigning a variable, remove inconsistent values from neighbours' domains; fail on empty.	Lo7 §4.6
Frequentism	Probability as long-run relative frequency over many independent trials.	Lo9a §3.1
Frontier	Set of generated-but-not-yet-expanded nodes; data-structure choice defines the strategy.	Lo3 §3.4
Fully observable vs partially observable	Whether sensors give the complete environment state at each time step.	Lo2 §3.6
Genetic algorithm (GA)	Population + selection + crossover + mutation, evolved over generations.	Lo5 §3.6
Gini impurity	$\text{Gini}(t) = 1 - \sum_j p(j \mid t)^2$; CART splitting criterion.	L10 §4.5
Goal-based agent	Agent maintaining explicit goal states and choosing actions toward them.	Lo2 §3.7
Goal state	The state(s) the agent is trying to reach; the goal test returns true here.	Lo3 §3.2
Heuristic function $h(n)$	Domain-specific estimate of cost from n to nearest goal; drives informed and local search.	Lo5 §3.1
Hidden Markov Model (HMM)	Hidden discrete Markov chain + observable emissions, parameterised by $\lambda = (A, B, \pi)$.	Lo9b §3.1
Hidden state	Latent unobserved state of the HMM; must be inferred from observations.	Lo9b §3.3

Term	One-line definition	Introduced in
Hierarchical clustering	Clustering producing a tree of nested clusters (agglomerative or divisive).	L12 §3.5
Hill climbing	Greedy local search: move to best-valued neighbour; stop at any local maximum.	L05 §3.3
Independence	$P(A \cap B) = P(A) P(B)$, equivalently $P(A \mid B) = P(A)$.	L09a §3.5
Inference by enumeration	Exact BN inference: sum the joint over atomic events consistent with evidence, normalise.	L09a §3.13
Information gain	Entropy(parent) – weighted average entropy of children; ID3/C4.5 split criterion.	L10 §4.6
Initial distribution (HMM)	Vector $\pi_i = P(q_1 = i)$; only multiplied in at $t = 1$.	L09b §3.2
Initial state	Starting state of the search (or empty assignment, in CSP).	L03 §3.2
Interaction term	Feature formed as product of two predictors; lets a model express varying slopes.	L11 §3.11
Intercept (regression)	Constant a in $y = a + bx$; baseline value when all predictors are zero.	L11 §3.2
Iterative deepening search (IDS)	Run depth-limited DFS for $d = 0, 1, 2, \dots$ until found; combines BFS optimality and DFS space.	L03 §4.4
Joint probability distribution	Probability over the full Cartesian product of variable domains; 2^n rows for n Booleans.	L09a §3.2
K-means clustering	Iterate: (1) assign each point to nearest centroid; (2) recompute centroids; until stable.	L12 §4.1
Learning agent	Agent with learning element + critic + performance element + problem generator.	L02 §3.7
Least Constraining Value (LCV)	CSP value-ordering: try the value that rules out the fewest options for remaining variables.	L07 §4.5
Linear regression	Fit $y = a + b_1 x_1 + \dots + b_p x_p$ minimising sum of squared residuals.	L11 §3.2
Local maximum	State whose value beats every neighbour but is below the global max — where hill climbing stops.	L05 §3.2
Local search	Keep one (or few) current states, iteratively modify; the goal is the value, not the path.	L05 §3.1
Marginal probability distribution	Distribution of a subset of variables; obtained by summing the joint over the others.	L09a §3.2

Term	One-line definition	Introduced in
Markov assumption (first-order)	$P(q_t \mid q_1, \dots, q_{t-1}) = P(q_t \mid q_{t-1})$.	Lo9b §3.2
Markov chain	States + transition matrix A + initial distribution; states are observable.	Lo9b §3.2
Markov condition (Bayes nets)	A BN node is conditionally independent of its non-descendants given its parents.	Lo9a §3.9
Minimax	Two-player zero-sum: MAX picks max-child, MIN picks min-child, recurse to terminals.	Lo6 §3.3
Minimum Remaining Values (MRV)	CSP variable-ordering: pick the variable with the smallest current legal domain — fail fast.	Lo7 §4.4
Model-based reflex agent	Reflex agent + internal state estimate + transition model — for partial observability.	Lo2 §3.7
Multi-agent	Environment with other agents whose actions affect the performance measure.	Lo2 §3.6
Multicollinearity	Two or more predictors strongly correlated — coefficients become unstable.	L11 §3.12
Mutation (GA)	Small-probability random bit-flip of an offspring; prevents premature convergence.	Lo5 §3.6
Naive Bayes classifier	$P(A_1, \dots, A_n \mid C) = \prod_i P(A_i \mid C)$; pick $\arg\max_C P(C) \prod_i P(A_i \mid C)$.	Lo9a §3.11
Node (search tree)	Data structure on the path; carries state, parent, action, path cost, depth. Distinct from "state".	Lo3 §3.3
Objective function	Real-valued function whose max (or min) defines the goal in local search / optimisation.	Lo5 §3.1
Observation (HMM)	Symbol emitted at a time step; the only information the inference algorithm sees.	Lo9b §3.3
Optimality	Algorithm guaranteed to return a least-cost solution whenever a solution exists.	Lo3 §3.6
Ordinary Least Squares (OLS)	Estimator: pick coefficients minimising $\sum_i (y_i - \hat{y}_i)^2$.	L11 §3.3
Overfitting	Model fits training noise — train error keeps falling, test error starts rising.	L10 §6.1
p-value	Probability of seeing a coefficient at least this extreme assuming the true coefficient is zero.	L11 §3.8
Path (search)	Sequence of actions/states from one state to another in the search graph.	Lo3 §3.2

Term	One-line definition	Introduced in
Path cost $g(n)$	Sum of step costs along a path; the optimal solution minimises path cost.	L03 §3.2
PEAS	Performance, Environment, Actuators, Sensors — four-line task-environment spec.	L02 §3.5
Percept	A single perceptual input at one moment — one sensor reading or one tuple of readings.	L02 §3.2
Percept sequence	The complete history of percepts up to the present moment.	L02 §3.2
Performance measure	External criterion (defined in terms of the environment) for judging the agent's behaviour.	L02 §3.3
Polynomial regression	Regression with $x^2, x^3, \dots$ added as features; flexible but overfit-prone. (FWD-REF — ML Lab 2.)	L11 (FWD-REF)
Population (GA)	The set of chromosomes considered together in one generation; typical 50–500.	L05 §3.6
Posterior probability	$P(X \mid e)$ — distribution after conditioning on evidence; output of probabilistic inference.	L09a §3.14
Prior probability	Unconditional distribution $P(X)$ before any evidence; root-node CPT in a BN.	L09a §3.14
Problem-solving agent	Goal-based agent that plans a sequence of actions via search through the state space.	L03 §3.1
R-squared (R^2)	$R^2 = \text{SSR} / \text{SST}$; proportion of total variance explained by the model.	L11 §3.6
Random forest	Ensemble of decision trees on bootstrap samples + random feature subsets at each split.	L10 §4.9
Random restart hill climbing	When hill climbing reaches a local max, restart from a random state; keep the best.	L05 §3.4
Random variable	Function from sample space to reals; mutually exclusive and exhaustive domains.	L09a §3.1
Rational agent	For every percept sequence, picks the action maximising expected performance given evidence.	L02 §3.3
Reflex agent (simple reflex)	Action depends only on current percept; condition-action rules; requires full observability.	L02 §3.7
Regression	Supervised learning with continuous-valued output (vs classification).	L10 §3.1
Reinforcement learning (RL)	Agent learns from delayed rewards; no labelled dataset, must explore.	L10 §3.1

Term	One-line definition	Introduced in
Residual (regression)	$r_i = y_i - \hat{y}_i$; OLS minimises $\sum r_i^2$.	L11 §3.3
Roulette-wheel selection	GA parent-selection where each chromosome's selection probability is proportional to fitness.	L05 §3.6
Search strategy	Rule for ordering frontier expansion — what distinguishes BFS / DFS / UCS / IDS / A*.	L03 §3.6
Search tree	Tree whose root is the initial state and children are states reachable in one action.	L03 §3.4
Simulated annealing	Local search that probabilistically accepts worsening moves with prob $\exp(-\Delta/T)$; T decays.	L05 §3.5
Single agent	Environment contains only one acting agent.	L02 §3.6
State	Representation of one configuration of the world relevant to the agent's reasoning.	L03 §3.3
State space	Set of all states reachable from the initial state by any action sequence.	L03 §3.3
Static vs dynamic	Whether the environment changes while the agent is deciding.	L02 §3.6
Stochastic hill climbing	Pick randomly among improving neighbours (or take the first improvement found).	L05 §3.4
Successor function	$\text{SUCCESS}(s)$: set of (action, resulting-state) pairs reachable from s .	L03 §3.2
Sum of squares (SST/SSE/SSR)	Total = Error + Regression: $\text{SST} = \sum (y_i - \bar{y})^2$, $\text{SSE} = \sum (y_i - \hat{y}_i)^2$, $\text{SSR} = \text{SST} - \text{SSE}$.	L11 §3.5
Supervised learning	Input/output pairs; model learns the mapping. Classification or regression.	L10 §3.2
Temperature schedule (SA)	$T(t)$: function decreasing temperature over time — linear, geometric, logarithmic.	L05 §3.5
Terminal state (game tree)	A state in which the game is over; carries a rule-given utility for MAX.	L06 §3.2
Test set	Held-out portion of the data used only to evaluate generalisation.	L10 §3.2
Training set	Data used to fit model parameters. Training error $\neq$ generalisation.	L10 §3.2
Transition model (Markov/HMM)	Matrix A with $a_{ij} = P(q_t = j \mid q_{t-1} = i)$; rows sum to 1.	L09b §3.2

Term	One-line definition	Introduced in
Transition model (search)	$\text{Result}(s, a)$: state that follows from action a in state s (deterministic counterpart).	L03 §3.2
Uncertainty	Property of environments where logical inference is insufficient — partial obs or stochastic.	L09a §3.1
Uniform-cost search (UCS)	Expand the lowest- $g(n)$ node (priority queue). Complete and optimal for non-negative costs.	L03 §4.2
Uninformed search	Search using only the problem definition — no heuristic estimates. BFS, DFS, UCS, IDS.	L03 §3.7
Unsupervised learning	No labelled outputs; algorithm discovers structure. Clustering, outliers, density estimation.	L10 §3.1
Utility-based agent	Agent using a utility function over states (real number, not yes/no goal).	L02 §3.7
Utility function	$U : S \rightarrow \mathbb{R}$; payoff at terminal states (games) or desirability per state.	L02 §3.3
Variable (CSP)	Symbol X_i that must be assigned exactly one value from its domain D_i .	L07 §3.1
Variable domain (CSP)	Set D_i of allowed values for X_i — finite, continuous, or numeric.	L07 §3.1
Viterbi algorithm	Like the forward algorithm but with $\max$ instead of $\sum$ and back-pointers — recovers the best path.	L09b §4.3
Zero-sum game	Two-player game where the sum of payoffs is constant — one player's gain is the other's loss.	L06 §3.5

139 glossary entries. Full definitions, notation, "alternative names seen in source", and lecture re-use history are in `_shared/glossary.md`.

6. Cross-Reference Graph

Which lecture introduces each concept (solid arrows) and which later lectures re-use it (dashed arrows). This graph is the *simplified* version of `_shared/cross-references.md`, focused on the load-bearing reuses; for the full edge list see that file.

```

graph LR
  L02["L02 Agents"]
  L03["L03 Uninformed Search"]
  L05["L05 Local Search"]
  L06["L06 Adversarial Search"]
  L07["L07 CSP"]
  L09a["L09a Bayesian Networks"]
  L09b["L09b HMM"]
  L10["L10 Intro to ML"]
  L11["L11 Regression"]
  L12["L12 Clustering"]

  %% L02 supplies the vocabulary every later lecture uses
  L02 -.->|agent, env types, PEAS, utility| L03
  L02 -.->|agent, multi-agent, utility| L06
  L02 -.->|rational agent, expected utility| L09a
  L02 -.->|partially observable, model-based| L09b
  L02 -.->|learning agent| L10

  %% L03 supplies the search-tree mechanics
  L03 -->|state, successor, frontier, DFS| L05
  L03 -->|game tree = search tree, DFS| L06
  L03 -->|backtracking is DFS over partial assignments| L07

  %% L05 forwards heuristic / objective function vocabulary
  L05 -.->|heuristic function = evaluation function| L06

  %% L07 lends "variable" vocabulary to probabilistic models
  L07 -.->|random variable ≈ CSP variable| L09a

  %% L09a → L09b and onward
  L09a -->|conditional independence, Markov condition, Bayes' rule| L09b
  L09a -->|Naive Bayes, Bayesian belief network| L10

  %% L09b feeds back the Markov assumption
  L09b -.->|Markov assumption| L09a

  %% L10 forks supervised / unsupervised paths
  L10 -->|regression task| L11
  L10 -->|clustering task| L12
  L10 -.->|RL evaluation function| L06

  %% L11 / L12 also reuse L10's vocabulary
  L11 -.->|overfitting| L10
  L12 -.->|distance / feature scaling| L11

  classDef lecture fill:#d9ead3,stroke:#274e13,stroke-width:1.5px,color:#000;
  class L02,L03,L05,L06,L07,L09a,L09b,L10,L11,L12 lecture;

```

How to read this. Solid arrows = the lecture *uses* the named concept set introduced by the source lecture. Dashed arrows = the lecture *references* concepts at a lighter touch (vocabulary reuse, motivating example). The full concept-by-concept graph lives in `_shared/cross-references.md` — that file has one explicit edge per glossary term.

7. Lab Reference Table

11 labs total — 8 algorithm labs (each a directory of `.py` files alongside a handout PDF) and 3 ML labs (Jupyter notebooks at the repo root). Each lab's entry-point solution file is what the Verifier executes end-to-end; the lab's variant bank is a list of exam-style perturbations you can drill against.

Lab	Entry-point solution	Handout
Lab 1 — Agents	<code>Lab1-Agents/reflex_vacuum_agent_solution.py</code> (+ <code>reflex_agent_with_state_solution.py</code> , <code>table_driven_agent_solution.py</code>)	<code>Lab 1.pdf</code>
Lab 2 — Search	<code>Lab 2/Search_solution.py</code>	<code>Lab 2/Lab 2.pdf</code>
Lab 4 — Genetic Algorithms	<code>handout_lab_4/ga_solution.py</code> (+ <code>Queen_solution.py</code> , <code>Number_solution.py</code> , <code>queens_fitness_solution.py</code>)	<code>handout_lab_4/Lab 4.pdf</code>
Lab 5 — Alpha-Beta / Tic-Tac-Toe	<code>handout/handout/tictactoe_template_solution.py</code> (uses <code>alpha_beta_solution.py</code>)	<code>handout/handout/Lab 5.pdf</code>
Lab 6 — CSP	<code>lab6/constraints_template_solution.py</code> (uses <code>States_solution.py</code> , <code>Colors_solution.py</code>)	<code>lab6/Lab 6.pdf</code>
Lab 7 — Bayesian Networks	<code>Lab7/handout/Runner_solution.py</code> (uses <code>bn_solution.py</code> , <code>Variable_solution.py</code>)	<code>Lab7/handout/Lab 7.pdf</code>
Lab 8 — HMM	<code>Lab 8/handout/hidden_markov_models_solution.py</code>	<code>Lab 8/handout/Lab 8.pdf</code>
ML Lab 1 — Classification	<code>lab1_classification_solution.ipynb</code>	(in-notebook)
ML Lab 2 — Regression	<code>lab2_regression_solution.ipynb</code>	(in-notebook)
ML Lab 3 — Clustering	<code>lab3_clustering_solution.ipynb</code>	(in-notebook)

Note on Lab numbering. There is no "Lab 3" in the AI series — the course skips from Lab 2 to Lab 4. The "ML Lab N" series is independent of the algorithm-lab numbering. Both series start at 1.

8. Common Pitfalls Compendium

Aggregated from every lecture's §6 "Common Pitfalls / Exam Traps". This is the single most exam-relevant page in this package after the analogies index. Drill it before the exam; if any row feels surprising, jump to the lecture's §6 section.

L02 — Agents

Pitfall	Remember this
"Intelligent agent always wins."	Rationality is <i>expected</i> performance — a rational agent can lose. Rationality $\neq$ omniscience.
Confusing agent function with agent program.	Function = abstract spec (potentially infinite); program = finite code realising it.
"Fully observable = agent knows everything."	It means <i>sensors</i> give the complete <i>environment</i> state — not knowledge outside the env (e.g. opponent psychology).
Episodic vs sequential confusion.	A sequence of moves inside one episode doesn't make the environment episodic. Episodic = choice in episode k has no effect on episode $k+1$.
"Static = nothing in the world moves."	Static = doesn't change <i>while the agent deliberates</i> . Chess-with-clock is <i>semi-dynamic</i> (perf measure time-dependent).
Discrete vs continuous as a whole-problem property.	Always specify the level of abstraction — chess board (discrete) vs chess robot arm (continuous).
"Multi-agent = >1 entity."	Other entities must <i>affect the agent's performance measure</i> to count.
PEAS items in the wrong column.	Sensors $\neq$ percepts (devices vs outputs). Actuators $\neq$ actions.
"Goal-based is smarter than reflex."	Architecturally yes, but reflex is faster in environments where simple rules suffice — match agent to env.
Performance measure = utility = evaluation function — always.	Three distinct things: performance measure (L02), expected utility (L09a), evaluation function (L06, non-terminal heuristic).
"Autonomous = moves by itself."	Autonomy = learning from experience, not movement.
Confusing one-shot randomness with stochastic transitions.	Only randomness in <i>transitions</i> makes env stochastic. Solitaire = deterministic after the deal; Scrabble = stochastic (ongoing draws).

L03 — Uninformed Search

Pitfall	Remember this
"Shallowest" vs "lowest cost".	BFS finds shallowest, <i>not</i> cheapest — when costs vary, BFS is not optimal.
Goal-test on generation vs expansion.	UCS <i>must</i> test on expansion (pop), not generation, to be optimal.
Forgetting completeness conditions.	BFS/IDS need finite b . UCS additionally needs step costs $\geq \epsilon > 0$. DFS not complete in infinite-depth/cyclic.
"DFS loops on cycles."	Pure DFS loops; DFS with the current-path check terminates but is incomplete on disconnected cyclic graphs.
State vs node confusion.	Multiple nodes can share a state. Keep the explored set keyed by <i>state</i> , not node-id.
UCS time = b^d .	No — UCS time is $b^{\lceil 1 + \frac{\log C^*}{\epsilon} \rceil}$; can be much worse than b^d on uneven costs.
"Fringe $\neq$ frontier."	Same data structure; slide name vs textbook name.
"IDS is slower than BFS."	Same big-O time $O(b^d)$; IDS uses DFS-style linear space.
"Lo3 covered A^* ."	It did not — slide 7 lists A as <i>objective</i> ; slide 54 says " <i>haven't covered</i> ". A is properly derived in Lo5.
d vs m .	d = shallowest goal depth (BFS/IDS); m = max path length in state space (DFS).

L05 — Local Search

Pitfall	Remember this
Sign-convention confusion.	8-puzzle uses negative objective (maximise $f \leq 0$); n-queens uses positive (minimise $h \geq 0$). State the sign at the top of your work.
"Hill climbing is complete."	False. Only on convex landscapes. Plain HC is not complete; SA is not "complete in practice"; GA has no formal completeness claim.
"SA is complete in practice."	The convergence proof needs $T = c/\log(1+t)$. Geometric schedules ($T \cdot 0.95$) lose the guarantee.
Plateau vs ridge vs local max.	Local max: strict decrease; HC stops. Plateau: equal neighbours; the slide-13 < rule loops indefinitely. Ridge: chained maxima; HC oscillates.
Skipping mutation in GA.	Without mutation, the population converges to identical chromosomes; crossover cannot reintroduce lost bits.
"Bigger population = better."	Half-true — more exploration but more cost. Slide-47 defaults ($N=50$, $m=0.05$, $c=0.9$) are starting points; tune empirically.
c vs m confusion.	c = per-pair crossover prob (~ 0.9); m = per-gene mutation prob (~ 0.05). Swap them and you get a random walk.
"Roulette will pick the best chromosome."	It picks proportionally — low-fitness can still win. Use <i>elitism</i> if you want the best preserved.
Genotype vs phenotype.	Genotype = the bitstring. Phenotype = what it decodes to. Fitness is on the phenotype.
Re-introducing the goal test in local search.	Local search has no goal test, only a termination condition (best fitness, step budget, $T \rightarrow 0$).

L06 — Adversarial Search

Pitfall	Remember this
Where α / β are updated vs used.	α updated at MAX, used at MIN; β updated at MIN, used at MAX.
Cutoff direction.	α is the floor (prune MIN when $v \leq \alpha$); β is the ceiling (prune MAX when $v \geq \beta$).
"Alpha-beta gives a different answer."	No — same minimax value as full minimax, only fewer nodes visited.
"Move ordering doesn't matter."	It does — best-first ordering gives $O(b^{d/2})$; worst-first degenerates to $O(b^d)$.
Eval at terminal states.	Eval is only for non-terminal cut-offs. Terminals use the rule-given utility.
Misreading "X's open lines – O's open lines".	Line is open for X if it contains <i>no</i> O (and vice versa). Empty lines are open for both and cancel.
"Minimax adapts to a weak opponent."	It doesn't — minimax plays as if the opponent is perfect. Actual payoff may exceed the minimax value.
Conflating eval function and utility function.	Utility = true terminal payoff. Eval = heuristic at non-terminal cutoffs.
Forgetting zero-sum precondition.	Minimax & alpha-beta as derived assume zero-sum. In non-zero-sum, MIN minimising MAX $\neq$ MIN maximising MIN.
Chess complexity.	$b \approx 35$, $d \approx 80$, $\approx 10^{123}$ nodes. (Not $b=80$, $d=35$.)

L07 — CSP

Pitfall	Remember this
Consistent $\neq$ complete.	Consistent: no constraint broken (may be partial). Complete: every variable assigned (may be inconsistent). Solution: both.
Variable assignment is commutative — but order matters for runtime.	Result is the same; backtrack count differs. MRV/degree exists for runtime, not correctness.
MRV vs degree.	MRV primary (fewest values left); degree is tie-breaker (most constraints on unassigned).
MRV vs LCV.	MRV chooses <i>variable</i> most likely to fail; LCV chooses <i>value</i> most likely to succeed. They cooperate.
Forward checking only looks at just-assigned variable's neighbours.	It does NOT propagate between two unassigned variables. Arc consistency does.
Arc consistency is sufficient.	It is necessary, not sufficient — you still need backtracking after AC.
Arcs are undirected.	They are <i>directed</i> . $X \rightarrow Y$ is not the same check as $Y \rightarrow X$. When Y shrinks, re-queue arcs <i>into</i> Y .
"Slide uses AC-3."	The slide procedure is AC-3 but the slide does not name it that. The textbook does.
"CSPs have no cost function."	Path cost is <i>constant per step</i> (slide 7), not absent.
"CSPs are polynomial because MRV/AC are."	They are NP-complete in general. Heuristics prune; worst case is still exponential.
Unary/global constraints in the binary constraint graph.	Unary fold into initial domains. Global (Alldiff) needs a hypergraph.
n-queens has two formulations.	Slide 12 (Boolean per cell) vs compact (one int per column). Different constraint graphs; specify which.

L09a — Bayesian Networks

Pitfall	Remember this
Mutually exclusive $\neq$ independent.	$A \cap B = \emptyset \Rightarrow$ anti-correlated, not independent.
Forgetting total-probability expansion in Bayes' rule denominator.	$P(B) = P(B \mid A) P(A) + P(B \mid \neg A) P(\neg A)$ \$. Skipping is off by a 5–10 $\times$ factor.
Base-rate fallacy.	A 50% sensitivity can give a 0.02% posterior when the prior is one-in-fifty-thousand.
Reading the wrong direction of a CPT.	$P(C \mid B)$ is <i>not</i> $P(B \mid C)$. Use Bayes' rule to flip; never just reverse the row.
Cycles in a BN.	A BN is a DAG. No cycles of any length. Cycles $\Rightarrow$ factorisation is not a probability distribution.
Independence $\neq$ conditional independence.	$X \perp Y$ does not imply $X \perp Y \mid Z$, nor vice versa.
Bad variable ordering when building a BN.	Order <i>causes before effects</i> . Bad order can force every later node to have many parents.
Forgetting root priors.	A root node carries a prior $P(X)$, not a CPT — but you still need it for factorisation.
" 2^{k+1} entries" rule.	That's table cells. Independent parameters are half (2^k) by row-sum-to-1.
Naive-Bayes zero-frequency.	One unseen feature value zeros out the whole product. Laplace smoothing fixes this.
"Conditioning always creates independence."	Wrong for colliders — conditioning on a common effect <i>creates</i> dependence (explaining away).
Forgetting to normalise after the joint sum.	Step 3 of inference by enumeration is division. If your conditional doesn't sum to 1, you skipped it.

L09b — HMM

Pitfall	Remember this
"Filtering" means different things.	Slides: $P(O \mid \lambda)$ (forward sum). Textbook: $P(q_t \mid o_1 \dots o_t)$ (normalised α).
Confusing α with Viterbi v .	Same trellis, same init $\pi_j b_j(o_1)$. Operator differs: $\sum$ vs $\max$.
Forgetting $b_j(o_t)$ at each cell.	Every cell ends "times $b_j(o_t)$ ". Common silent error.
a_{ij} vs a_{ji} .	Convention: $a_{ij} = P(q_t = j \mid q_{t-1} = i)$. Row = source, column = destination.
Multiplying by π at every step.	π_j is the start probability — only used at $t = 1$.
Brute force vs DP complexity.	$O(N^2 T)$ enumeration vs $O(N T)$ forward/Viterbi.
Markov assumption vs output-independence assumption.	Different assumptions. HMM uses both.
Forgetting back-pointers in Viterbi.	The $\max$ gives you the score; back-pointers give you the <i>sequence</i> .
Rows vs columns sum to 1.	A : rows sum to 1 (must transition somewhere). B : rows sum to 1 (must emit something).
Stationarity.	A, B don't change with time in standard HMM. Time-varying transitions $\Rightarrow$ not standard.
Numerical underflow.	Long sequences underflow — work in log-space.
"Three problems list is exhaustive."	Problem 3 (Learning / Baum-Welch) is named, not derived, in this course.

L10 — Intro to ML

Pitfall	Remember this
Underfitting vs overfitting.	Underfit: train and test both bad. Overfit: train great, test bad. Test error has a U-shape.
Two causes of overfitting.	Cause 1: noise (mislabelled points). Cause 2: insufficient examples (sparse regions inherit boundaries). Different fixes.
Confusing train and test error.	Training error always falls with capacity. Only test error tells you about generalisation.
Computing Gini with wrong probabilities.	$p(j \mid t)$ is the relative frequency <i>at node t</i> , not the global class prior.
Forgetting weighting in $\text{Gini}_{\text{split}}$.	It is the <i>weighted average</i> over children — a pure child of size 1 doesn't outweigh a noisy child of size 100.
"More than one tree fits = bug."	Fundamental: different splitting orders give different trees on the same data.
Confusing entropy and information gain.	Entropy = single-node impurity. Info gain = parent entropy – weighted child entropy. ID3 uses info gain.
Too-deep trees for tiny datasets.	Cause 2 of overfitting: in tiny regions the tree picks up noise from irrelevant nearby points.

L11 — Regression

Pitfall	Remember this
Reading R^2 without context.	$R^2 = 0.25$ is great for social science, dismal for chemistry. Always frame by domain.
Statistical vs practical significance.	Tiny coef + p $\approx 0 \neq$ practically important. Always quote the confidence interval too.
Comparing raw R^2 across models with different predictors.	Raw R^2 never decreases when you add a variable. Use adjusted R^2 for model comparison.
Dummy-variable trap.	Categorical with k levels needs $k - 1$ dummies. Adding the k -th creates perfect multicollinearity.
Forgetting the baseline category.	The intercept is the value when <i>all dummies</i> = 0. Other-category effects are <i>relative to the baseline</i> .
Sign of a dummy after adding an interaction.	The "main effect" of the dummy with an interaction present is the effect <i>at zero of the interacted variable</i> , not on average.
Extrapolating beyond the data range.	The intercept at $x=0$ may be far outside your training range.
Multicollinearity $\rightarrow$ unstable coefs, not bad predictions.	Predictions may still be accurate; individual coefficients are not interpretable.
Linearity is always appropriate.	Saturation / inverse-U effects need non-linear features (polynomial, log, interactions).
Confusing SST/SSE/SSR names.	Total / Error / Regression. $SST = SSE + SSR$. $R^2 = \frac{\text{SSR}}{\text{SST}}$.
Per-predictor p-values vs F-statistic.	Per-coef p = "does <i>this</i> predictor matter given the others?" F p = "does <i>any</i> predictor matter?"
L11 does not name MSE / RMSE / gradient descent.	Those are ML Lab 2 territory. Don't quote them as L11 concepts on an exam.
Intercept p-value.	Tests "intercept = 0?" — usually irrelevant. Ignore unless the business question is specifically about zero.
Misstating the p-value's meaning.	Slide 19's "probability that the predictor has no effect" is the inverse fallacy. Correct: probability of seeing a coefficient this extreme <i>assuming</i> the true coef is zero.
$R^2 \in [0, 1]$ always.	Only on training data with intercept. On held-out test or no-intercept fit, R^2 can be negative .
Forgetting units.	Slope 7.527 is "per unit of Advertising", which may be $\$1000$ of ad spend, not $\$1$.

L12 — Clustering

Pitfall	Remember this
K-means finds the optimum.	It converges to a local minimum. Re-run from many inits; report SSE.
Convergence = correct answer.	Convergence just means centroids stopped moving. Different inits give different converged answers.
Choosing K by eyeballing.	The lecture does not prescribe a method. Lab-3 answer: elbow or silhouette.
Treating the dendrogram as canonical.	Different linkages (MIN, MAX, average, centroid, Ward) give different dendrograms on the same data. State the linkage first.
Once-merged-never-undone.	Hierarchical agglomerative is greedy. A wrong early merge is permanent.
MIN-link chaining.	A bridge of intermediate points fuses visually distinct clusters into one. Avoid MIN if data has noise between clusters.
DBSCAN parameter sensitivity.	Tiny <code>Eps</code> changes flip results. Use k-distance plot. Fails on varying-density data — OPTICS is the textbook follow-up.
Noise vs outlier in K-means.	K-means has no noise concept; every point joins some cluster. Use DBSCAN (or K-medoids) for noisy data.
High-dimensional curse.	Distance concentrates $\Rightarrow$ "nearest neighbour" loses meaning. Reduce dimensions (PCA) first.
Forgetting feature scaling.	Euclidean is dominated by the largest-range feature. Always z-score (or min-max) when features have heterogeneous units.
"Validation gives a definitive answer."	Slide 44: clusters are in the eye of the beholder; validation is the most difficult and frustrating part of cluster analysis.
Topics this lecture omits.	K-means++, elbow, silhouette, OPTICS, GMM/EM, spectral clustering — none on the slides. They are ML Lab 3 territory.

End of Master Index. Total: 10 lectures · 73 analogies · 139 glossary terms · 11 labs · 125+ pitfalls. If you have read every row of §4 and §8, you are well-prepared. Good luck.